Chapter 8 — Plant Physiology

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Stomata open when guard cells: (a) Lose K+ (b) Close aquaporins (c) Take up K+ (d) Release ABA
2. ABA closes stomata by: (a) Opening aquaporins (b) Sugar formation (c) More K+ influx (d) Blocking K+ uptake
3. TACT mechanism explains: (a) Mineral uptake (b) Stomatal movement (c) Food translocation (d) Ascent of sap
4. Casparian strip blocks apoplast in: (a) Endodermis (b) Cortex (c) Pith (d) Epidermis
5. Phloem source is where: (a) Water lost (b) Minerals absorbed (c) Sugar consumed (d) Sugar produced/loaded
6. Apical meristems cause: (a) Secondary growth (b) Fruit ripening (c) Primary growth (d) Bark formation
7. Annual rings form by seasonal activity of: (a) Cambium (b) Epidermis (c) Stomata (d) Root cap
8. Vernalisation is flowering induced by: (a) Touch (b) Long days (c) High temperature (d) Low temperature

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. How do guard cells control the opening and closing of stomata?
2. Explain the TACT mechanism of water transport in plants.
3. How does the pressure-flow theory explain translocation of food?
4. Differentiate between apical and lateral meristems.
5. How are annual rings formed in trees?
6. How does phytochrome help plants measure the length of darkness?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Describe the mechanisms of stomatal opening and closing. **(4)**
(b) Explain the uptake and transport of water in plants. **(4)**
Q.4 (a) Describe the structure of phloem and the pressure-flow theory of translocation. **(4)**
(b) Explain photoperiodism and vernalisation. **(4)**

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